

Fabricating multilayer antireflective coating for near complete transmittance in broadband visible light spectrum

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ABSTRACT

To extract maximum performance out of the certain optical devices it is highly desirable to use completely transparent window in a broad wavelength range. In search of large-scale applicable antireflective (AR) coating for such window, herein we report the design and fabrication of multilayer coating with the aim to reduce the reflection of the window in the entire visible light range. Systematic selection of material and thickness of each layer was performed based on the criteria fulfilled by refractive index to obtain zero reflection. Low refractive index material like MgF_2 is deposited by e-beam technique to obtain single-layer AR coating. While Al_2O_3 and ZrO_2 , deposited by RF-magnetron sputtering, with MgF_2 are used to develop multilayer AR coating. Scanning electron microscope image along with elemental line scan obtained along the cross-section of the coating verified the thickness and composition of each layer. Deposition of four-layer AR coating ($\text{MgF}_2/\text{Al}_2\text{O}_3/\text{ZrO}_2/\text{MgF}_2$) on both side of quartz window with optimized thickness showed considerable reduction in the reflection to 0.6% at 550 nm. The average transmittance of 99.1% was recorded in the broadband of visible spectrum in the range of 450–750 nm wavelength. With near to complete transmission, the present AR coating holds encouraging prospect for implementing in windows of digital display, photovoltaic module, and other optoelectronic devices.

1. Introduction

Ever increasing demand of optical and optoelectronic devices has driven the scientific community to identify the effective strategies to maximize light transmission and receiving efficiency. In this context, protective window with high optical transparency plays a pivotal role. For example the ultimate efficiency of photovoltaic module will depend on the reflection property of the cover glass from where minimum of 4% of incident light across the solar spectrum is reflected back and lost. Space exploration through telescope receives a very minute optical signal which cannot be afforded to be attenuated through the window. Among others, reflection losses through the lenses of cameras and microscopes affect the quality of images. Generally, antireflective (AR) coatings and pattern on the windows and lenses are used to avoid such shortfall [1,2]. In addition to accomplish minimum reflection, AR coating also need to be effective in broadband spectrum, independent on the angle of incidence, and should be ultra thin [3,4].

Single-layer AR coating, with refractive index value between that of air and of the window material, and with thickness of one quarter of

wavelength of interest, reduces the reflection at that wavelength [5]. However, practically such material does not exist in the nature thus MgF_2 having low refractive index close to the ideal value is used for single layer coating. Materials with high porosity and low density were also applied for such single-layer AR coating with limited success [6–8]. Combination of two materials with different refractive index has also been tested in two-layer AR coating which manages to reduce the reflection of only specific wavelength [2,9]. In order to further broaden the bandwidth of the spectrum, multilayer AR coatings have been realized from materials having refractive index greater than that of the window [10–16]. By proper selection of the material with required refractive index and tuning the thickness, the multilayer coating can minimize the summation of overall reflection vectors [17]. The combination of TiO_2 and SiO_2 is most commonly used for multilayer AR coating [18–22], while Al_2O_3 having refractive index of 1.62 close to glass is rarely been studied along with ZrO_2 [9,14,23]. Both these materials have been limited to the designing stage [9,23] even if refractive index of both these materials perfectly satisfy the condition of refractive index required to achieve zero reflection for multilayer coating.

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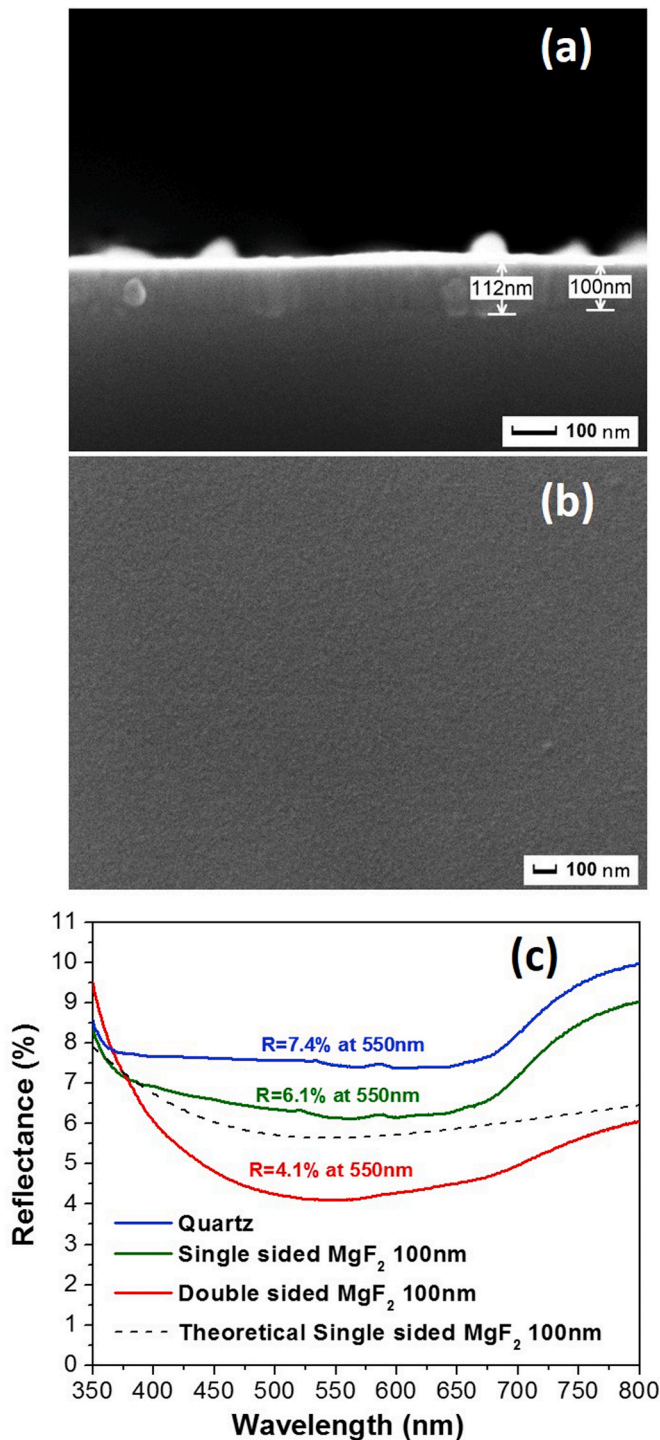


Fig. 1. (a) Cross-section, and (b) surface SEM micrographs, along with (c) reflectance spectra of single-layer (Q/MgF₂) AR coating deposited on single and double side of quartz substrate.

2. Experimental methods

Electron-beam deposition (EBD) and RF-magnetron sputtering deposition techniques were used to synthesize single and multilayer AR-coatings. Single layer of MgF₂ was deposited by EBD on polished and cleaned quartz substrate. Prior to deposition the chamber was pumped to a base pressure of 10^{-6} mbar. MgF₂ film was deposited with the rate of 2–3 nm/min and the film thickness was controlled by thickness monitor to achieve final thickness of 100 nm. Pure ZrO₂ (99.9%) and Al₂O₃

(99.9%) disc (10 cm) was used as the target to deposit multilayer AR-coatings by RF-magnetron sputtering. High vacuum with base pressure less than 3×10^{-7} mbar was achieved before deposition while the working Ar pressure of 10^{-2} mbar was used for the deposition. Before sample deposition, both oxide targets were pre-sputtered for 20 min in order to remove any surface contamination. ZrO₂ and Al₂O₃ targets were sputtered on cleaned surface of both quartz and Si (100) substrates at room temperature using RF power of 150 W. The sample and target distance was maintained at 6 cm for all the deposition. Sputtering time was adjusted and calibrated to attain the required thickness of the layers. **Two-layer AR-coating** was synthesized by first depositing Al₂O₃ thin film by sputtering and then transferring to the EBD chamber for MgF₂ thin film deposition on top of Al₂O₃ layer. **Three-layer AR-coating** was synthesized by depositing ZrO₂ thin film on top of Al₂O₃ layer by sputtering and then depositing MgF₂ thin film by EBD. Additional layer of MgF₂ was first deposited on the substrate, by EBD, before depositing above mentioned three layers stacks (Al₂O₃/ZrO₂/MgF₂) to achieve **four-layer AR-coating**. The thickness of each layer in the multilayer coating was selected after an appropriate calculation to achieve broad band antireflection. We paid maximum attention to the stoichiometry, purity, and homogeneity of all the materials produced, precisely to be sure that the optical properties of the prepared materials were similar to those of the literature reported materials. Indeed, as reported below, the theoretical reflectance calculated using Fresnel equation are in well agreement with all our experimental results.

The performance of different AR-coatings was tested by measuring percentage reflectance in the visible light range (300–800 nm) of the spectrum. Optical measurements were performed in a **Bruker IFS66 spectrometer** in a reflection mode with incidence angle of 15° for incoming beam. TFCalc TM design software was used to obtain theoretical reflectance plot for designed single and multilayer antireflection coating at angle of incidence of 15°. The surface morphology of the samples was analyzed by using scanning electron microscope (SEM, JEOL) equipped with Energy Dispersive Spectroscopy (EDS) through which compositional studies of the samples were performed.

3. Results and discussion

3.1. Single layer AR-coating

For single layer AR-coating, MgF₂ with refractive index (refractive index = 1.38) close to the required value ($n_1=1.21$) of a single layer (accordingly to $n_1=\sqrt{n_0n_s}$) [5] was selected, considering the RI of air ($n_0=1$) and quartz substrate ($n_s=1.47$). Since the thickness of the AR-coating plays a key role in diminishing the reflection, thus in present case the thickness of MgF₂ layer was selected as 100 nm such that it obeys quarter-wavelength ($\lambda/4n_1$) criteria for the chosen reference wavelength of 550 nm. In order to reduce the overall reflection losses, MgF₂ layer was deposited on top as well as on bottom side of the quartz substrate. MgF₂ thin film was deposited by EBD technique on quartz substrate with a thickness of ~100–110 nm as confirmed by cross-section SEM image (Fig. 1a). The layer was compact and dense as noticed along the cross-section (Fig. 1a), while the top view (Fig. 1b) of the film shows that surface is smooth and flat without any major defects. Fig. 1c presents the theoretically plotted and experimentally measured reflectance spectra, in the visible light range, of the bare quartz substrate and with MgF₂ AR-coating. The percentage reflectance of ~7.5% is detected at wavelength of 550 nm for quartz substrate and this value is consistent with the theoretical value obtained after reflection from both the side of the quartz substrate [24]. The reflectance value of quartz substrate decreases to ~6.1% at 550 nm by depositing MgF₂ layer on the single side which further decreases to ~4.1% for the double sided AR coating. This shows that it is necessary to coat the quartz substrate on both the sides to decrease the reflection.

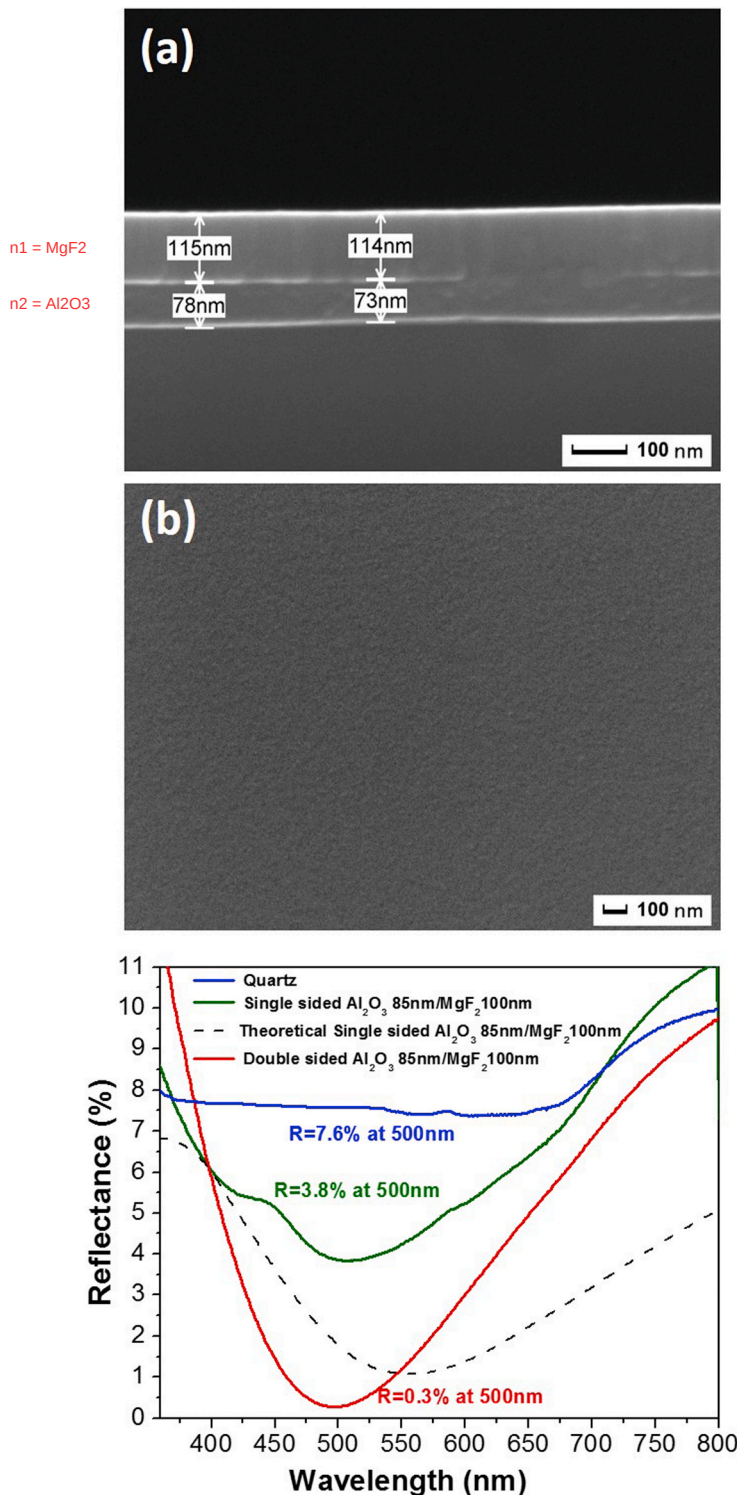


Fig. 2. (a) Cross-section, and (b) surface SEM micrographs, along with (c) reflectance spectra of two-layer (Q/Al₂O₃/MgF₂) AR coating deposited on single and double side of quartz substrate.

3.2. Two layer AR-coating

The **two-layer quarter/quarter AR coating** is used as an alternative to the single-layer coating exhibiting limited reduction in the reflectance. **Owing to higher refractive index of MgF₂ than required**, it results in stronger reflection from the first surface (air/MgF₂) which is not completely cancelled through destructive interference with the weaker

reflection from the second surface (MgF₂/quartz). On the other hand, in case of two-layer AR-coating two weaker reflections, generated from underlying surfaces, combine to cancel out-of-phase reflection from the top surface. The necessary condition to select the refractive index of the two-layer AR coating with quarter-quarter wavelength thickness is [1]:

$$\frac{n_1}{n_2} = \sqrt{\frac{n_0}{n_s}}$$

By selecting **MgF₂** as the top layer, to fulfill this condition **Al₂O₃** with refractive index (n_2) of 1.62 was deposited on quartz substrate by RF-magnetron sputtering prior to the deposition of MgF₂ layer. The thickness of MgF₂ (~100 nm) and Al₂O₃ (~85 nm) was around one-quarter ($\lambda/4n$) of the wavelength (550 nm) of interest as confirmed by cross-section SEM image (Fig. 2a) which also illustrates the compact and dense nature of both Al₂O₃ and MgF₂ layers. Similar to single layer, the top surface of MgF₂ appears even and defect-less (Fig. 2b). V-shaped reflectance spectrum (Fig. 2c) is observed for the single-sided two-layer (MgF₂/Al₂O₃) AR-coating with minimum reflectance value of 3.8% recorded at a wavelength of 500 nm. The V-shape reflectance is consistent with the reported two-layer AR-coating of different materials where the range of valley of V-shape depends on the thickness and refractive index of two layers. Hence, in the present case the lowest reflectance value is shifted to 500 nm from the expected 550 nm, due to the lower thickness value (73–78 nm) achieved for Al₂O₃ layer than the calculated (~85 nm). Nevertheless, significant decrease in reflectance value of mere 0.3% at 500 nm was detected when two-layer coating is deposited on the both side of quartz substrate. This shows that **two-layer AR coating** is able to reduce the reflectance of quartz substrate with higher efficiency as compared to single-layer. However, two-layer AR coating is useful for the near monochromatic applications because it shows minimum reflectance at the designed wavelength such as 500–550 nm.

3.3. Multilayer AR Coating

Broadband AR coating was designed by involving three-layer coating with **quarter-half-quarter wavelength thickness**. Such layers were designed in order to meet the condition of refractive index in form of $n_{top}^2 n_s = n_{bot}^2$ [19] to obtain zero reflection at a certain range of wavelength, where n_{top} and n_{bot} are the refractive index of top and bottom layer respectively. With quartz substrate and top MgF₂ layer, ideal refractive index for bottom layer is calculated to be about 1.68. Thus, **Al₂O₃** (refractive index = 1.62) was selected for the bottom layer. The above equation also reveals that refractive index of middle layer has no effect on the condition to record zero reflectance at wavelength of interest. Hence, middle layer is also called as **absentee layer** as the half wavelength thickness of this layer do not affect the overall performance at certain wavelength. Nevertheless, **refractive index of the middle layer decides the range of broadband along which the reflection will be minimum**. According to the simulation performed through software (TFCalc) the middle layer with refractive index = 2.2 fits perfectly with the top MgF₂ and bottom Al₂O₃ layer to show the average transmittance of about 99% in broad 450–600 nm range (Fig. 3d). Thus **ZrO₂** with refractive index = 2.14 was selected as the middle layer and deposited by RF-magnetron sputtering with a half wavelength thickness ($\lambda/2n = 128$ nm). Although through cross-section SEM image (Fig. 3a) the total thickness of three-layer coating of ~310 nm was verified, however it was difficult to distinguish each layer in the thin film stack. Thus to view three layers, a compositional line scan along the cross section was performed and presented in Fig. 3b which endorses that ZrO₂ was evidently sandwiched between the MgF₂ and Al₂O₃. Smooth top surface of MgF₂ was illustrated in Fig. 3c. The reflectance spectra (Fig. 3d) of the three layer AR coating (Quartz/Al₂O₃/ZrO₂/MgF₂) on single side of quartz causes the percentage reflectance to decrease in broad-band range (500–800 nm) of visible light spectrum with a value of about 3.1% at

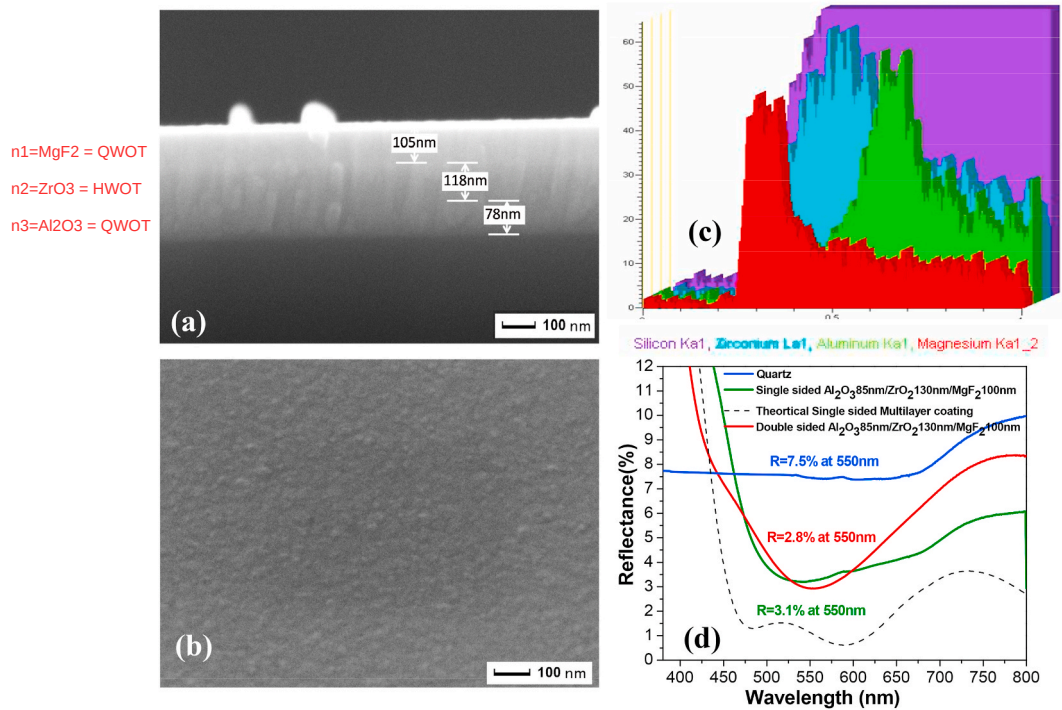


Fig. 3. (a) Cross-section and (b) surface, SEM micrographs, (c) Composition analysis along the cross-section, and (d) Reflectance Spectra of the **three-layer (Q/Al₂O₃/ZrO₂/MgF₂)** AR coating deposited on single and double side of quartz substrate.

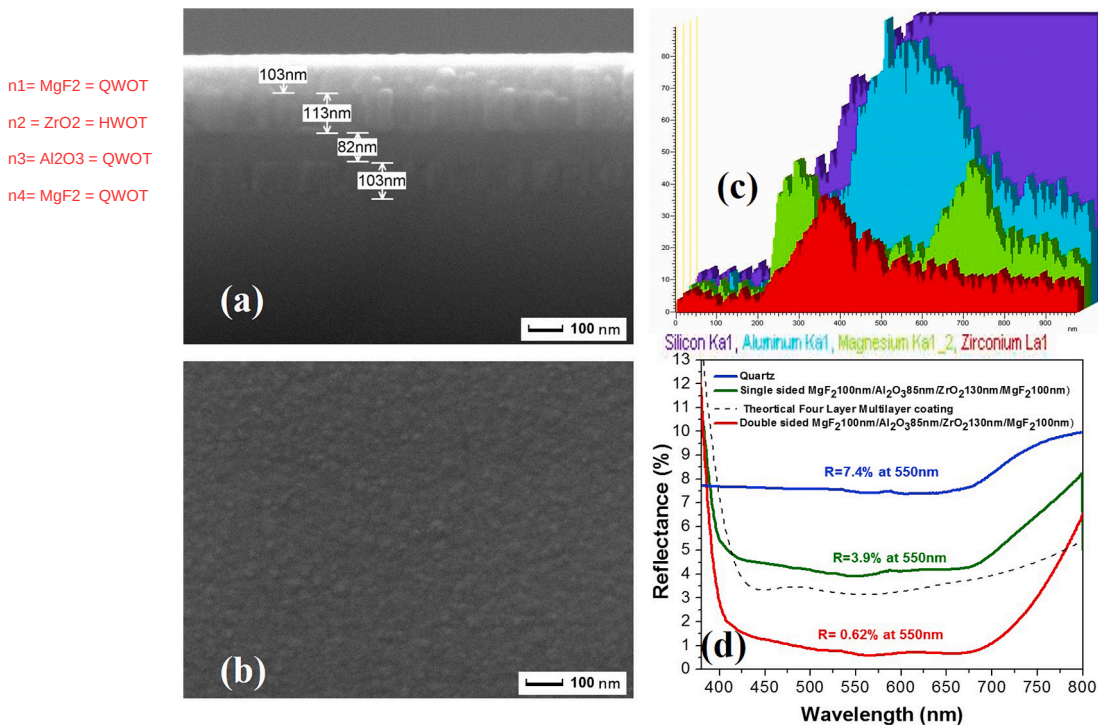


Fig. 4. (a) Cross-section and (b) surface, SEM micrographs, (c) Composition analysis along the cross-section, and (d) Reflectance Spectra of the **four-layer (Q/MgF₂/Al₂O₃/ZrO₂/MgF₂)** AR coating deposited on single and double side of quartz substrate.

550 nm. Surprisingly, **the double-side coating** of these layers do not cause the further decrease in the reflectance value as observed in single-layer and two-layer AR coating. Even with single-side coating, **the antireflection effect in three-layer AR coating is constrained in a narrow range** of visible light spectrum with cutoff wavelength near 500 nm below which the reflection increases drastically to a very high value.

In order to solve the above problem **four-layer AR coating** was designed by introducing an extra layer adjacent to quartz substrate in the three-layer configuration. The fourth layer was selected based on the criteria of $n_1 n_4 = n_3$ [25], with known value of n_1 (MgF₂) and n_3 (Al₂O₃), the n_4 is calculated to be around 1.17. Since ideally no material exists with such refractive index, so again MgF₂ with quarter wavelength

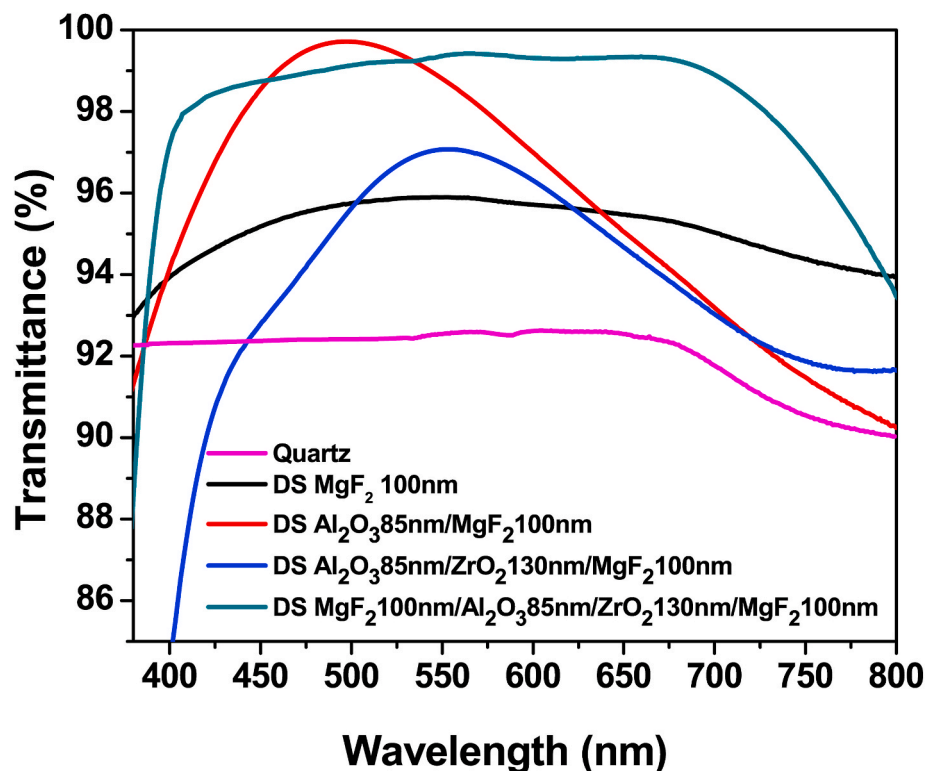


Fig. 5. Comparison of Transmittance Spectra of all AR coatings deposited on double side of quartz substrate.

thickness was deposited on the quartz substrate prior to the three-layer AR coating. The cross-section SEM image (Fig. 4a) along with compositional lines scan (Fig. 4b) displays all four layers with $\text{Al}_2\text{O}_3/\text{ZrO}_2$ layers sandwiched between MgF_2 layers. The flat nature of the top MgF_2 surface seems to be maintained even in this type of coating (Fig. 4c). By coating **four-layers on single-side of quartz** causes the percentage reflectance to decrease in complete broad-band range of 400–750 nm in visible light spectrum with a value of $\sim 3.9\%$ at 550 nm (Fig. 4d). By depositing **double-sided four-layer AR coating** the percentage reflectance decreases further to a value of $\sim 0.6\%$ in complete broadband range of visible light. As depicted by this result, the four-layer AR coating is the most effective in decreasing the reflection of the quartz.

Comparison plot of percentage transmittance of all the AR coatings deposited on double-side of quartz substrate is reported in Fig. 5. Average percentage transmittance in the visible light range (450–750 nm) increases to 95.5% and 96.5% from $\sim 92.2\%$ of quartz substrate by depositing single-layer and two-layer AR-coating, respectively. In case of multilayer coating, even if ordinary improvement is registered with three-layer AR coating ($\sim 94.9\%$), the maximum transmittance of 99.1% was attained by four-layer AR coating in region of interest. The above results show that systematic selection of the material based on their refractive index and fine tuning their thickness can be very useful strategy to achieve near about zero reflection from the “transparent” window used in various optical and optoelectronics applications.

4. Conclusion

In summary, **the single-layer, two-layer and multilayer AR coating** was deposited on the quartz window by physical deposition techniques. The selection of materials as well as the thickness of the layers was based on the criteria fulfilled by the relationship of refractive index. **Percentage reflectance significantly decreased from 7.5% to 0.6% at 550 nm wavelength** by depositing four-layer AR coating ($\text{Quartz}/\text{MgF}_2/\text{Al}_2\text{O}_3/\text{ZrO}_2/\text{MgF}_2$) on double-side of quartz window with the thickness of quarter-quarter-half-quarter wavelength. This form of AR coating

allowed nearly complete transmission of the visible light by showing average transmittance as high as $\sim 99.1\%$ in broadband of 450–750 nm. The present multilayer AR coating is simple to design based on the condition of refractive index and can be fabricated on large scale by industrial deposition techniques. With high performance and practical applicability, this AR coating seems to be a promising candidate for transparent window for optical devices including photovoltaic.

CRediT authorship contribution statement

Rupali Patel: Methodology, Data curation, Software, Investigation. **Nainesh Patel:** Conceptualization, Writing - original draft. **Nicola Bazzanella:** Methodology, Investigation. **Antonio Miotello:** Supervision, Writing - review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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